# RAAJ VAMSY RAYAVARAPU

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**Principal Engineer — React Native Platform & Applied AI**

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Hyderabad, India

## SUMMARY

Principal Engineer on the WaveMaker React Native platform, promoted from Senior SDE. Shares architecture and technical direction with the founding architect and team leads. Owns the developer-facing stack: **wm-reactnative-cli**, **Appchef**, **Expo** integration, and **Web Preview**.

Independently built and shipped **MemoryLayer** (MCP-native memory for AI agents) and **ChefQuest** (AI recipe app). Active R&D in **RAG**, **computer vision**, **algorithmic trading**, and **cognitive architecture**.

## SKILLS

| | |
|---|---|
| **Languages** | Python · TypeScript / JavaScript · Java · C · C++ · C# · GDScript |
| **Mobile & Web** | React Native · Expo · Next.js · Angular · Spring Boot · FastAPI / Flask · Supabase · PostgreSQL · MySQL |
| **AI & Data** | LLM APIs (OpenAI, Anthropic, Google Gemini) · Ollama · embeddings · FAISS · ChromaDB · RAG and multi-agent retrieval · MCP (tool design, stdio and HTTP transports) · prompt orchestration · OpenCV, OCR-style pipelines, CLIP-style matching · Z3 formal verification |
| **Infrastructure** | Docker · Elasticsearch / Kibana · SQLite |
| **R&D / Personal** | Godot 4 · Blender (rigging, animation, Python API) · Qt |

## EXPERIENCE

### Principal Engineer — WaveMaker, Hyderabad

Apr 2025 – Present

- **Agentic engineering pipeline.** Built an end-to-end pipeline from an issue ticket to a buildable reproduction project, automating spec generation, codebase reingestion, test generation (functional, edge-case, security, performance) and code generation. It cut bug and feature turnaround substantially, and the QA team has since adopted the same approach.
- **Image-to-code.** Originated and built the pipeline that turns a screenshot into platform markup, then compares a re-render against the source. It became a **standard way apps are built on the platform**, and the web and mobile teams now each maintain their own version on this base.
- **Runtime performance.** Profiled and refactored the React Native runtime to reduce component render time across apps built on the platform.
- **Preview architecture.** Re-architected the web preview pipeline around a shared transform cache and build admission control, significantly improving repeat-build time, peak memory, live-reload speed and build reliability.
- **Build observability and AI root-cause analysis.** Built a platform that makes preview-build data trustworthy, groups failures into error codes by owning side, runs an AI agent to write root-cause reports, and routes alerts to the right people. Engineers can query build data in plain English through **MCP**.
- **Performance diagnostics.** Built the base toolkit for diagnosing React Native app performance: it captures runtime events and syncs them with screen recordings in a timeline report showing time to first interaction, API time and re-render cost.
- **Platform ownership.** Own **wm-reactnative-cli**, **Appchef**, **Web Preview** and **Expo** integration; built the **Expo Router** migration across codegen and runtime, and **MCP integrations** for the platform.

### Senior Software Development Engineer — WaveMaker, Hyderabad

May 2022 – Apr 2025

- **Founding engineer.** Rejoined as a founding member of the new **WaveMaker Mobile (React Native)** platform team and built the codebase from inception; early on, also covered QA and direct enterprise customer support, including onsite visits, until a dedicated QA lead took over.
- **Platform ownership.** Owned **wm-reactnative-cli**, **Appchef** (build pipeline), **Expo** integration, and **Web Preview** end to end — releases, mobile preview, CLI behavior, and release-blocking issues for customers and the services organization.
- **Product engineering & R&D.** Shipped features, bug fixes, and architectural changes across the React Native platform, and led exploratory R&D / POC efforts spanning Expo, bundling, and platform internals.
- **Engineering standards.** Set the code review and integration bar for large parts of the codebase; partnered with professional services, QA, and enterprise accounts on upgrades, defects, and performance.
- **Developer documentation.** Led the **WaveMaker Learn** upgrade to **Docusaurus v3** with **Algolia** search.

### Freelance Software Developer — Self-employed, Remote

Dec 2020 – Mar 2022

- Built **Flask**, **Django**, and **Spring Boot** backends, **microservices**, scraping and automation tools, and **OpenCV**-based utilities for independent clients.

### Software Developer — Pramati Technologies (WaveMaker), Hyderabad

Apr 2019 – Nov 2020

- Built **mobile (Cordova)** and **web applications** on the **WaveMaker low-code platform** for enterprise customers, after starting on internal tooling.
- Delivered work directly for enterprise customers, including a large international scientific organisation.
- Extended and supported the Java-based WaveMaker IDE and platform using **Spring Boot**, **Angular**, and **MySQL**.

### Software Trainee — Ingram Micro, Chennai

Jul 2018 – Dec 2018

- Built and maintained warehouse desktop software with **C#**, **ASP.NET MVC**, **Angular**, and **MySQL**.

### Freelance Software Developer — Self-employed, Remote

Mar 2017 – Jul 2018

- Shipped **Qt** desktop applications and automation tools for independent clients.

## PRODUCTS

### MemoryLayer memorylayer.in

Founder; in production since May 2026. Local-first, **MCP-native** persistent memory for AI agents. Combines hybrid **vector/BM25** retrieval with a **Z3-based formal verification layer** that catches agent behavioral drift. Built and launched solo: architecture, implementation, release engineering, and go-to-market (npm, MCP Registry, Show HN).

### ChefQuest chefquest.in

Founder, since Nov 2025. Generative-AI recipe product built on **Next.js** and **Supabase**.

### Plantopia

Gamified AI plant-care app with camera capture, weather data and growth timelines, built on **Next.js**, **Supabase** and **Llama 4 vision**.

### Hush

**Android** privacy app built on **Expo** and a native **VpnService**: blocks internet per app, shows which hosts each app contacts, and identifies and silences unwanted calls and SMS.

## INDEPENDENT R&D

**68+ personal repositories** over the last two years, mostly private; details available on request. Highlights:

- **FFIC and FunnelZip (lossless compression, C):** FFIC image coder produces files **1.5–4.9% smaller than JPEG XL at effort 9** on 20-photo and 4K sets, **5–6.7% smaller** on 16-bit elevation data, and beats FFV1, x264 and x265 lossless on RGB test clips, at much slower speed. General-purpose FunnelZip beat the best of xz, brotli, zstd, FLAC, PNG and WebP lossless on **25 of 27 files over 10 KB**. All benchmarked output decoded exactly.
- **Artificial organism research:** one body-independent learning core (**world-model ensemble**, **episodic memory**, imagination, habit) that learns from instincts and its own choices. Tested on three simulated bodies with **pre-registered protocols**: criteria met on one, partly met on a second, and not yet on the third.
- **Algorithmic trading:** Binance futures market-making pipeline with a **Z3 formal risk gate** and walk-forward backtesting that caught overfitting and reported "no validated edge."
- **Cognitive architecture (FunnelBrain):** non-LLM, physics-based "wave-field" model of language and reasoning, with each hypothesis documented as supported or refuted.
- **RAG and agents:** code-intelligence stacks (FAISS, graph retrieval, multi-agent debate); **godot-mcp** editor automation; **aiflow** LLM workflow builder on React Flow.
- **Games and vision:** Godot games (**Icy Tower** shipped), image retrieval, and image-to-3D Blender pipelines.

## EDUCATION

### B.E., Computer Science and Engineering — Sathyabama University, Chennai — Graduated 2018